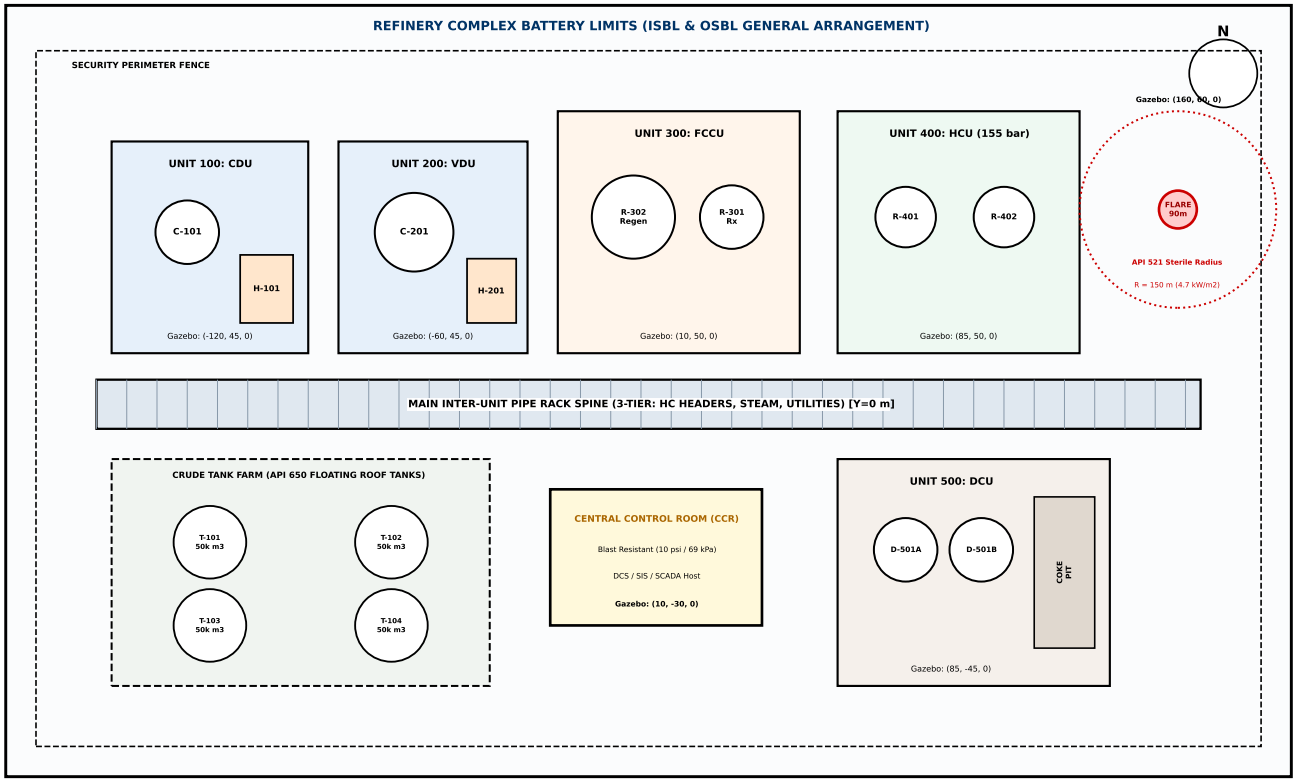
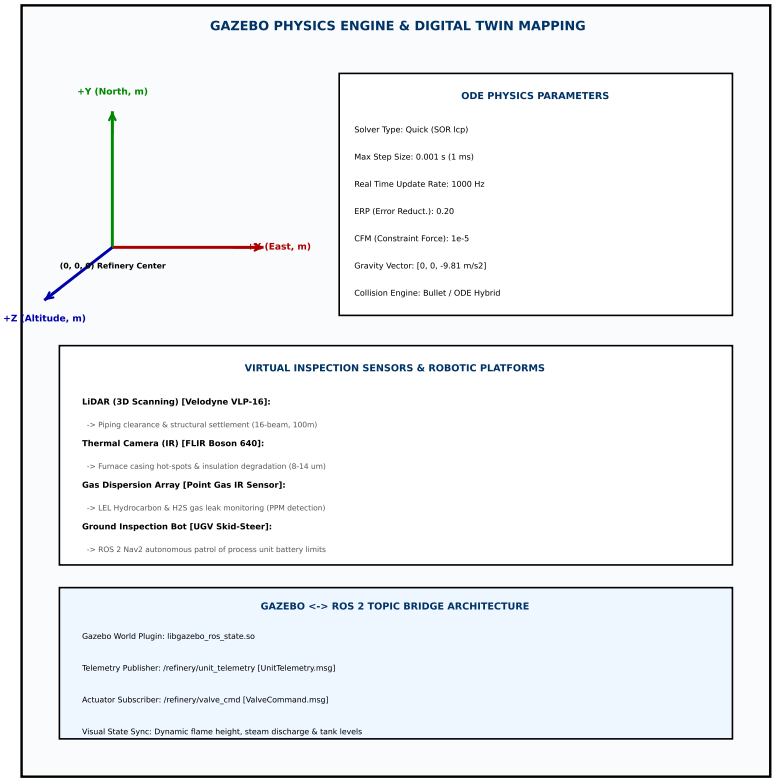
OVERALL REFINERY PLOT PLAN, BATTERY LIMITS & GAZEBO 3D SPATIAL COORDINATES

SCALE: 600 m x 400 m PLANT FOOTPRINT | API RP 500 / NFPA 30 / OISD-118 SAFETY SPACING | ODE PHYSICS INTEGRATION





REFINERY BATTERY LIMITS, GAZEBO SPATIAL LAYOUT & INDUSTRIAL SAFETY SPACING STANDARDS

EQUIPMENT / UNIT BLOCK	GAZEBO COORD [X, Y, Z] (m)	DIMENSIONS (LxWxH m)	MIN SPACING CODE	SAFETY CRITERIA & HAZARD CLASSIFICATION
Unit 100: Atmospheric CDU (C-101)	[-120.0, 45.0, 0.0]	60.0 x 50.0 x 62.0	API RP 500 / OISD	Fired heater H-101 separated 15 m from column shell
Unit 200: Vacuum VDU (C-201)	[-60.0, 45.0, 0.0]	50.0 x 45.0 x 48.0	API RP 500 / OISD	Vacuum ejector barometric seal leg & sour water sump
Unit 300: FCCU Riser / Regen	[10.0, 50.0, 0.0]	70.0 x 60.0 x 65.0	OISD-118 Cl 5.2	Catalyst hopper separation; CO boiler flue ducting
Unit 400: Hydrocracker Complex	[85.0, 50.0, 0.0]	80.0 x 65.0 x 42.0	NFPA 30 Class I	High-pressure H2 zone (155 bar); 30 m blast buffer
Unit 500: Delayed Coker (Drums A/B)	[85.0, -45.0, 0.0]	75.0 x 55.0 x 92.0	OISD-118 Cl 6.4	Coke pit deluge system; unheading blast shield separation
Crude Tank Farm (T-101 to T-104)	[-100.0, -50.0, 0.0]	140.0 x 90.0 x 22.0	NFPA 30 / API 650	Bund dyke capacity > 110% largest tank volume
Elevated Flare Stack (API 521)	[160.0, 60.0, 0.0]	Dia 1.8 m x H 90.0 m	API 521 Sec 5.4	Sterile radius R=150 m (thermal radiation < 4.7 kW/m2)
Central Control Room (CCR)	[10.0, -30.0, 0.0]	40.0 x 25.0 x 8.0	API RP 752 / 753	Blast-resistant design (69 kPa); positive pressurization

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS PLANT DIGITAL TWIN SIMULATION

DRAWING TITLE	LAB / SUBJECT	COURSE / BRANCH	SCALE
REFINERY PLANT PLOT PLAN & GAZEBO 3D LAYOUT	**PROCESS PLANT DIGITAL TWIN & CAD**	**B.E. CHEMICAL ENGINEERING**	**1 : 2500 (METRIC)**
STUDENT NAME	**ROLL NO.**	**DRAWING NO.**	**DATE**
SOURISH SENAPATI	**002310301035**	**JU/ChE/VP/DWG-04**	**19 / 10 / 2026**

VIRTUAL PETROLEUM REFINERY | 100,000 BPD CRUDE DIET | GAZEBO & ROS 2 DIGITAL TWIN | API / ASME BPVC